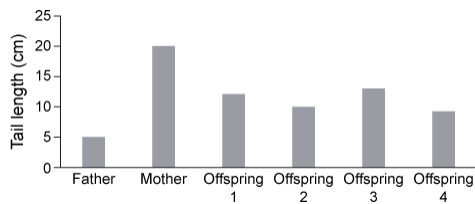


A gene regulating tail length in bobtail cats has a genetic variation at a single locus, where organisms can have an allele that produces long tails (L) or an allele that produces short tails (l). A male cat homozygous for short tail length (ll) was mated with a female cat homozygous for long tail length (LL). The tail lengths of the parents and offspring are recorded below.



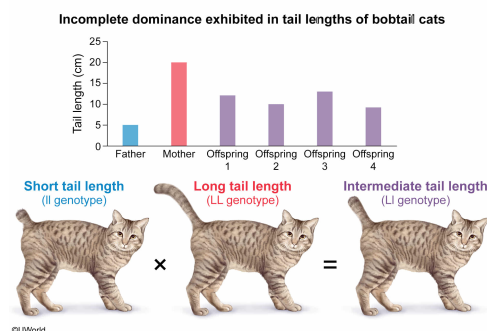
A scientist concludes that the alleles regulating tail length in these cats exhibit incomplete dominance. Do the data support this conclusion?

- ☐ A. No, because all offspring have tail lengths longer than the father's tail. [2%]
- ☐ B. No, because all offspring inherited both types of alleles for tail length. [3%]
- ☐ C. Yes, because the tail lengths of some offspring are longer than the tail lengths of other offspring. [6%]
- ☒ D. Yes, because the offspring have tail lengths longer than the father's tail but shorter than the mother's tail. [87%]

Correct

87% answered correctly

Explanation:



Individuals **heterozygous** for a trait possess two distinct alleles for the gene encoding that trait. However, the way that these distinct alleles interact to give rise to specific traits in a heterozygous individual can vary. In some heterozygotes, a *completely dominant* allele in one copy of the gene **phenotypically masks** a recessive allele in the other copy of the gene. In other cases, **both alleles** can be phenotypically expressed, producing heterozygous individuals with combined or **blended traits**.

Specifically, **incomplete dominance** between alleles results in a blended or **intermediate** phenotype in heterozygous individuals. A common example of incomplete dominance is a cross between homozygous **red** flowers (RR) and homozygous **white** flowers (WW) producing heterozygous offspring (RW) that are **pink**.

In the given scenario, a gene regulating tail length in bobtail cats has two alleles at the same locus: an allele producing long tails (L) and an allele producing short tails (l). Mating a male cat homozygous for short tail length (ll) with a female cat homozygous for long tail length (LL) produces **heterozygous offspring** (Ll) possessing *both* alleles. The graph shows that these heterozygous offspring have tail lengths longer than the father's but shorter tails than the mother's; therefore, these **offspring have an intermediate phenotype**. This observation *supports* the conclusion that these alleles **exhibit incomplete dominance**.

(Choice A) Although all the heterozygous offspring have tail lengths longer than the father's, they all also have *shorter* tail lengths than the mother's. Therefore, this observation *supports* the conclusion that the alleles influencing tail length in these cats exhibit incomplete dominance. In contrast, if inheritance were *completely* dominant/recessive, all offspring tails should be the *same length* as one of the parents.

(Choice B) Although heterozygotes inherit both types of alleles, only observations of phenotypic *traits* in these individuals inform scientists of whether the alleles exhibit incomplete dominance or another mechanism of inheritance (eg, completely dominant/recessive inheritance, *codominance*). For example, codominant inheritance requires offspring to express *both* a short *and* a long tail length instead of a single, intermediate-length tail.

(Choice C) Although the tail lengths *among* offspring do vary, this observation alone indicates variable *expressivity* (*not* incomplete dominance) of the phenotype. Determination of incomplete dominance inheritance requires that the tail lengths of the offspring be *compared with those of the parents*.

Educational objective:

Incomplete dominance between alleles results in the expression of phenotypes in heterozygous offspring that are intermediate to the phenotypes of homozygous parents.

Time spent:0 sec

Subject:

Biology

QID:402428

System:

Genetics and Evolution